

Generalization of the Matrix Limit

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We can define a generalized theorem for an $n \times n$ matrix whose elements all share a common leading-order linear term in their Maclaurin series expansions.

1. Theorem Statement

Let $A(x)$ be an $n \times n$ matrix where each entry $a_{ij}(x)$ is an analytic function in the neighborhood of $x = 0$.

Suppose every entry has a Maclaurin series expansion of the form:

$$a_{ij}(x) = kx + c_{ij}x^p + O(x^q)$$

where $k \neq 0$ is a common constant across all entries, $p > 1$ is the first non-uniform higher-order power, and $q > p$.

Then, there exists a finite, non-zero limit for the determinant scaled by a specific polynomial degree, given by:

$$\lim_{x \rightarrow 0} \frac{\det(A(x))}{x^{p(n-1)+1}} = k \sum_{i=1}^n \sum_{j=1}^n \text{adj}(C)_{ij}$$

where C is the $n \times n$ coefficient matrix of the x^p terms (i.e., $C = [c_{ij}]$), and $\text{adj}(C)$ is its adjugate matrix.

2. Rigorous Proof

We can express the generalized matrix $A(x)$ algebraically by separating the linear terms and the higher-order terms:

$$A(x) = kxJ + x^pC + O(x^q)$$

where $J = \mathbf{1}\mathbf{1}^T$ is the $n \times n$ matrix of all ones.

To evaluate the determinant of $A(x)$, we invoke the Matrix Determinant Lemma, which describes the determinant of a matrix subjected to a rank-1 update ($\det(M + uv^T) = \det(M) + v^T \text{adj}(M)u$). Here, our base matrix is $M(x) = x^pC + O(x^q)$, and our rank-1 update is kxJ :

- **Step 1:** Apply the identity to $A(x)$:

$$\det(A(x)) = \det(x^pC + O(x^q)) + kx\mathbf{1}^T \text{adj}(x^pC + O(x^q))\mathbf{1}$$

- **Step 2:** Extract the leading-order terms from both components using the properties $\det(cxM) = (cx)^n \det(M)$ and $\text{adj}(cxM) = (cx)^{n-1} \text{adj}(M)$:

$$\det(A(x)) = x^{pn} \det(C) + kx \cdot x^{p(n-1)} \mathbf{1}^T \text{adj}(C)\mathbf{1} + \dots$$

- **Step 3:** Simplify the expression to find the dominant term as $x \rightarrow 0$:

$$\det(A(x)) = x^{pn} \det(C) + kx^{p(n-1)+1} \sum_{i,j} \text{adj}(C)_{ij}$$

- **Step 4:** Compare the powers of x . Because $p > 1$, we have $pn > p(n-1) + 1$. Thus, the lowest power of x (and therefore the dominant term as $x \rightarrow 0$) is exactly $x^{p(n-1)+1}$.

Consequently, dividing the determinant by $x^{p(n-1)+1}$ eliminates the x dependency of the lowest order term, sending all higher order terms (like x^{pn}) to 0, leaving only the summation of the adjugate matrix.

3. Application to the Provided Image

We can directly map the theorem to the visual problem to prove why x^7 is in the denominator.

- **Matrix Dimension (n):** The image displays a 3×3 matrix, so $n = 3$.
- **Common Linear Term (k):** The Maclaurin series for all entries ($\tan(x)$, $\sin(x)$, x , $\sinh(x)$, etc.) begins with $1 \cdot x$, so $k = 1$.
- **Next Non-Zero Power (p):** For these specific trigonometric and hyperbolic functions, the x^2 term is 0, and the next coefficient appears at the cubic term x^3 . Thus, $p = 3$.

Using our generalized power formula $p(n - 1) + 1$:

$$3(3 - 1) + 1 = 3(2) + 1 = 7$$

This confirms that the determinant of this matrix grows proportionally to x^7 as x approaches 0. The limit effectively extracts the sum of the cofactors of the x^3 coefficient matrix C .